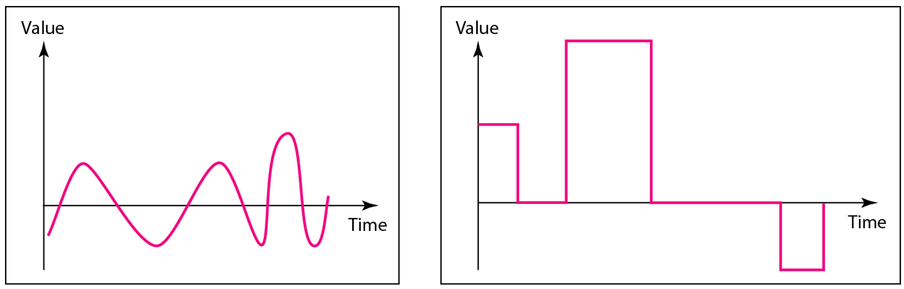
02 The physical layer

02.01 Basic principles of transmissions

Analog and Digital

- Data can be analog or digital
  - Analog data refers to information that is continuous
  - Analog data take on continuous values
  - Analog signals can have an infinite number of values in a range
  - Digital data refers to information that has discrete states
  - Digital data take on discrete values
  - Digital signals can have only a limited number of values

Comparison of analog and digital signals

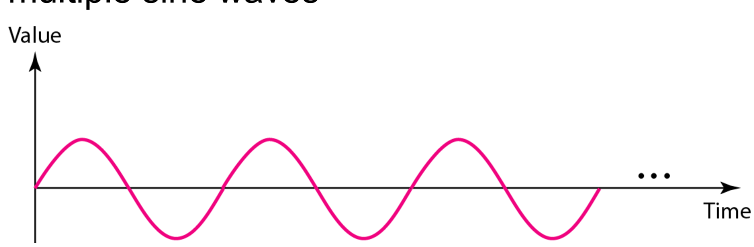


a. Analog signal b. Digital signal

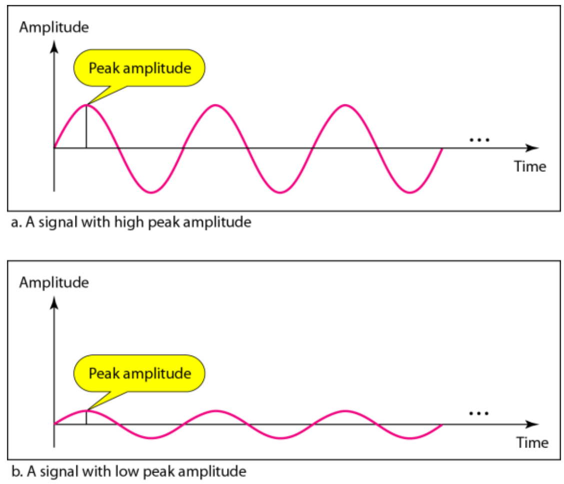
- In data communications, we commonly use periodic analog signals and non periodic digital signals.

Periodic analog signals

- Periodic analog signals can be classified as simple or composite
- A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals
- A composite periodic analog signal is composed of multiple sine waves



Signal Amplitude



Frequency

- Frequency is the rate of change with respect to time
- Change in a short span of time means high frequency.
- Change over a long span of time means low frequency.
- If a signal does not change at all, its frequency is zero
- If a signal changes instantaneously, its frequency is infinite.

Frequency and Period

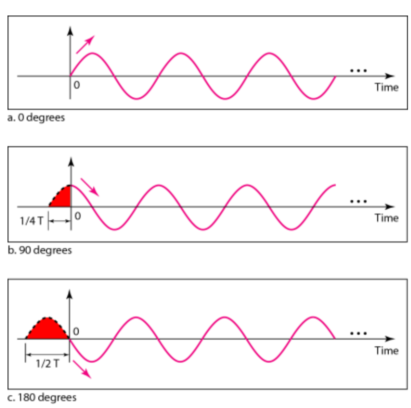
- Frequency and period are the inverse of each other

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

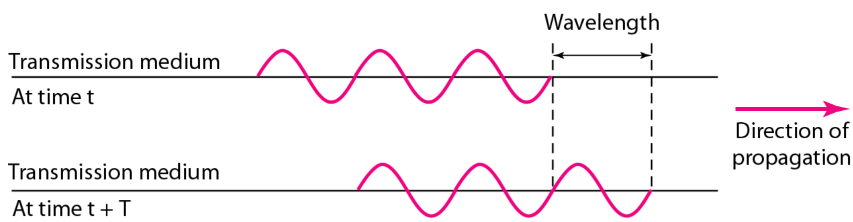
Phase

- Phase describes the position of the waveform relative to time 0



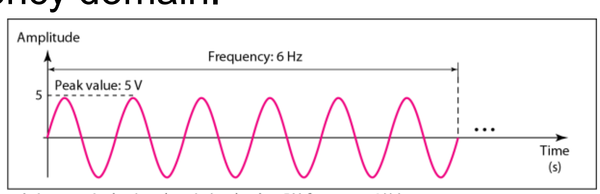
Wavelength and period

- Wavelength = Propagation speed x Period
  = Propagation speed / Frequency

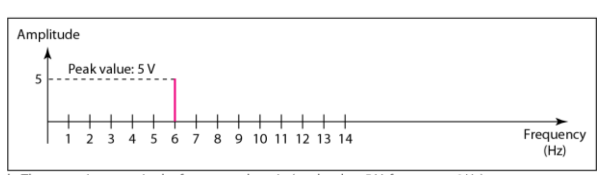


Time-domain and frequency-domain

- A complete sine wave in the time domain can be represented by one single spike in the frequency domain.

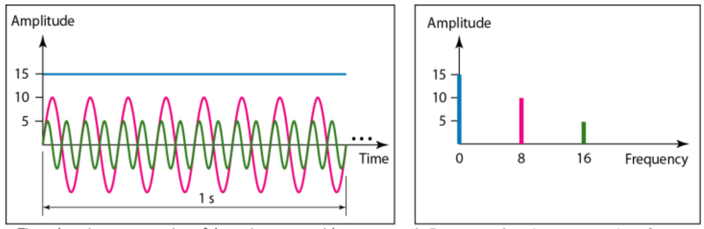


a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Frequency Domain



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16

b. Frequency-domain representation of the same three signals

- The frequency domain is more compact and useful when we are dealing with more than one sine wave.
- A single-frequency sine wave is not useful in data communication
  - We need to send a composite signal, a signal made of many simple sine waves.

Fourier analysis

- According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.
  – If the composite signal is periodic, the decomposition gives a series of signals with discrete frequencies;
  – If the composite signal is nonperiodic, the decomposition gives a combination of sine waves with continuous frequencies.

Fourier series expansion

$$g(t) = \frac{1}{2} \cdot c + \sum_{n=1}^{\infty} a_n \cdot sen(2\pi nft) + \sum_{n=1}^{\infty} b_n \cdot \cos(2\pi nft) \qquad (1)$$

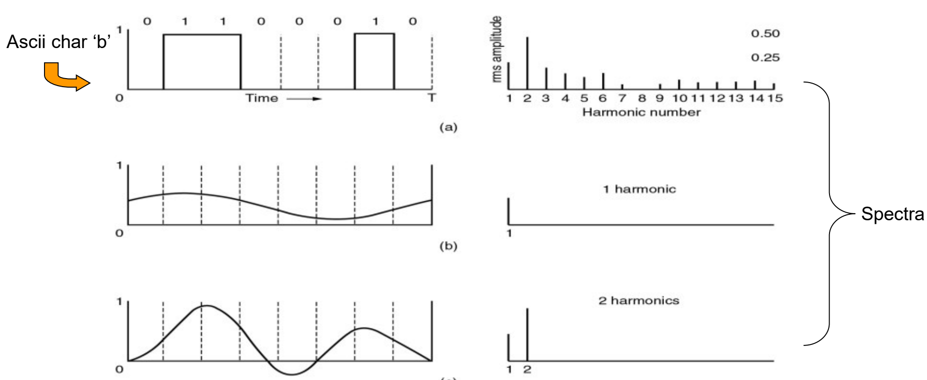
$$\int_0^T sen(2\pi kft) sen(2\pi nft) dt = \begin{cases} 0 \; per k \neq n \\ T/2 \; per k = n \end{cases} \qquad (2)$$

$$a_n = \frac{2}{T} \int_0^T g(t) sen(2\pi nft) dt \qquad (3.a)$$

$$b_n = \frac{2}{T} \int_0^T g(t) \cos(2\pi nft) dt \qquad (3.b)$$

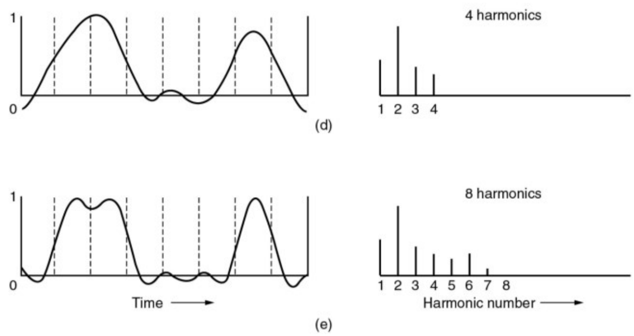
$$c = \frac{2}{T} \int_0^T g(t) dt \qquad (3.c)$$

Bandwidth-Limited Signals



- A binary signal and its root-mean-square Fourier amplitudes. (b) – (c) Successive approximations to the original signal.
- → $rms = \sqrt{a_n^2 + b_n^2}$
- $(rms)^2$ α (transmitted energy of the corresponding frequency)

Bandwidth-Limited Signals



- Successive approximations to the original signal (d) – (e)

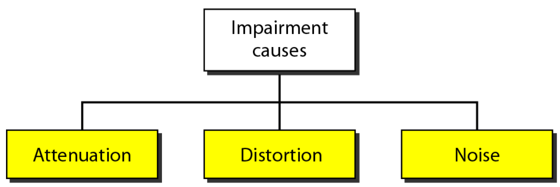
Bandwidth-Limited Signals

Bps	T (msec)	First harmonic (Hz)	# Harmonics sent
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

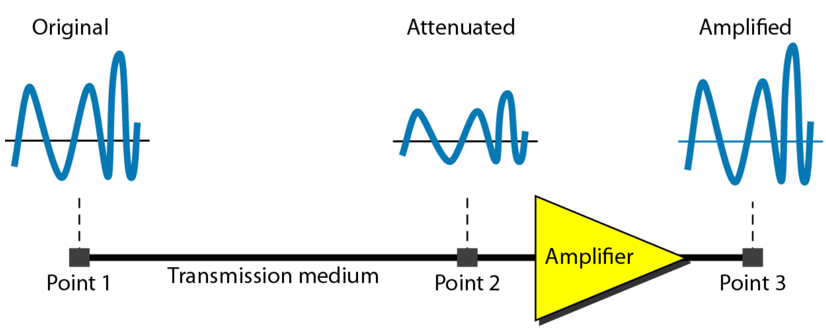
Relation between data rate and harmonics on a 3000 Hz channel

TRANSMISSION IMPAIRMENT

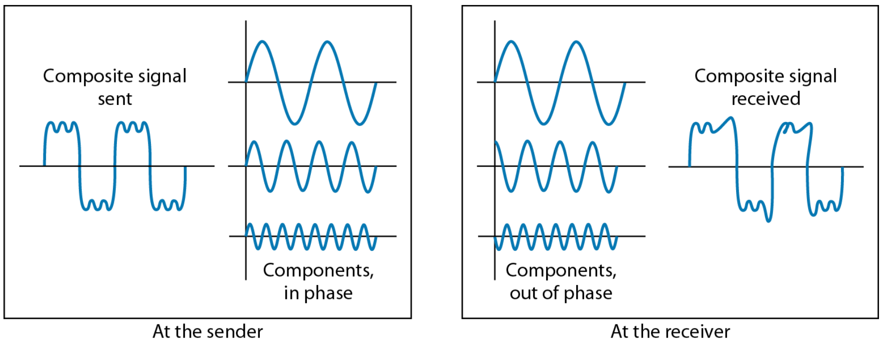
- Signals travel through transmission media, which are not perfect.
- The imperfection causes signal impairment.
- This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.
- What is sent is not what is received.
- Three causes of impairment are **attenuation, distortion**, and **noise**.



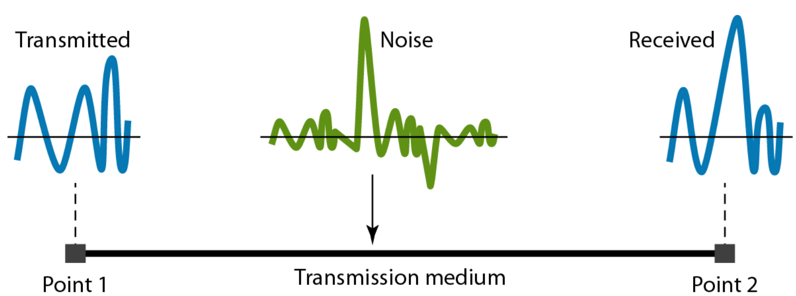
Attenuation



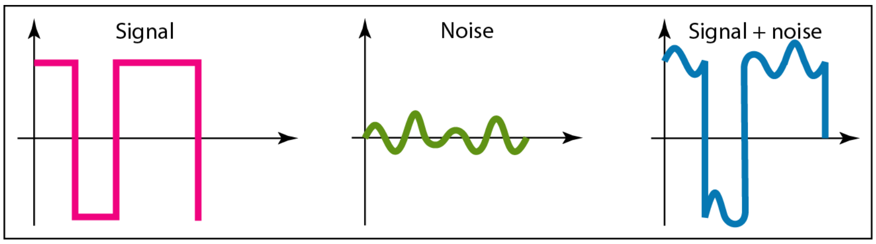
Distortion



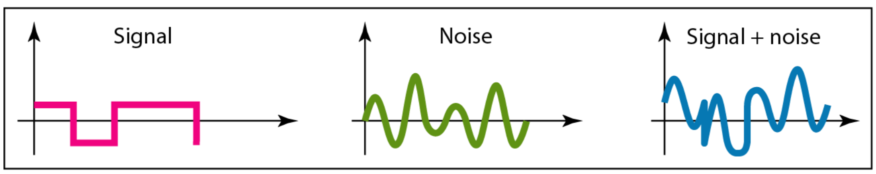
Noise



Two cases of SNR



a. Large SNR



b. Small SNR

Nyquist and Shannon theorems

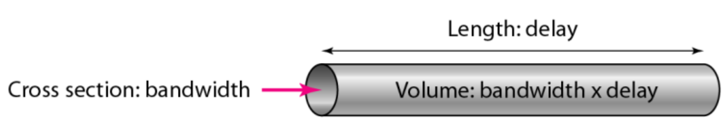
- Nyquist:
  - A signal sent over channel with bandwidth *H* can be correctly reconstructed by taking *2H* samples/sec. If the signal has *V* discrete levels →
    - ***max bit rate=2H*** $\boldsymbol{\log_2 V}$ ***(bit/sec)***
    - e.g. a noiseless 3 kHz channel supports, for binary signals, up to 6000 bps
- Shannon:
  - Nyquist theorem doesn't take into account thermal noise.
  - S = signal power, N = noise power, S/N=Signal to Noise Ratio ($10\log_{10} S/N$ (dB)) →
    - ***max bit rate=H*** $\boldsymbol{\log_2 (1+S/N)}$ ***(bit/sec)***
    - e.g. a noisy 3 kHz channel with S/N = 30 dB, supports, independently from the number of levels and the sampling frequency, up to 30000 bps

Performance

- One important issue in networking is the performance of the network—how good is it?
- In networking, we use the term bandwidth in two contexts
- The first, bandwidth in **hertz**, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- The second, bandwidth in **bits per second**, refers to the speed of bit transmission in a channel or link.

Bandwidth-delay product

- The bandwidth-delay product defines the number of bits that can fill the link.



We can think about the link between two points as a pipe.

The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay.

We can say the volume of the pipe defines the bandwidth-delay product.